3. Background and Preminilaries

3.1 Selection of Red Channel for Secure Data Embedding

A digital colour picture consisting of three colour planes, red, green, and blue [19, 24]. The RGB picture can be expressed as shown in equation (1):

$$P_{RGB}(x,y) = \{R(x,y), G(x,y), B(x,y)\} \quad (1)$$

where $R(x,y)$, $G(x,y)$, and $B(x,y)$ signifies the red component, green component, and blue component intensity values for the pixel coordinate (x,y).
The RGB picture can also be represented as a 3D matrix [22][25] as shown by equation (2):

$$P_{RGB} \in \mathbb{R}^{M\times N\times 3} \quad (2)$$

where M and N denotes the picture height and width, respectively.

The individual colour bands are segregated as shown in equation (3) for R, equation (4) for G and equation (5) for B channel as below:

$$R = P_{RGB}(:,:,1) \quad (3)$$
$$G = P_{RGB}(:,:,2) \quad (4)$$
$$B = P_{RGB}(:,:,3) \quad (5)$$

In the proposed strategy, only the red channel is considered for secret information embedding and frequency-domain processing [16]. Therefore, the obtained embedding domain is represented by equation (6) as follows:

$$P_R(x,y) = R(x,y) \quad (6)$$

The energy of the selected red domain is calculated by equation (7) as:

$$E_R = \sum_{x=1}^{M}\sum_{y=1}^{N} R^2(x,y) \quad (7)$$

where E_R represented as total energy of the signal contained by red component.
The proportional contribution of the red channel to RGB picture can be evaluated by equation (8) as:

$$\eta_R = \frac{E_R}{E_R + E_G + E_B} \quad (8)$$

where E_G and E_B denotes the energies of the respective green and blue sub channels, respectively.
Transformation operations are executed only on the red channel for decreasing the computational complexity. The computational proportional reduction can be approximately evaluated by equation (9) as follows:

$$C_R = \frac{1}{3} C_{RGB} \quad (9)$$

where C_{RGB} signifies the cost of processing for analysis of all three channels and C_R denotes the red domain cost for the same.
The designated red channel is then forwarded to the hybrid transform phase following equation (10) given below:

$$P_R \overset{\text{DWT-DCT}}{\rightarrow} F_R \quad (10)$$

where F_R represented as frequency channel coefficients utilized for secret information embedding strategy.

3.2 Discrete Wavelet Transform (DWT) Method

DWT as shown in Figure 1 is quite effective for multi-resolution analysis where image is divided into different frequency sub-bands. This capability makes it an essential part of method that enables analysis at different resolutions. By encoding data in higher-frequency sub-bands it is possible to satisfy requirements like robustness [26] as well as invisibility. This approach helps in identifying another solution that is best in meeting the need for robustness and transparency.

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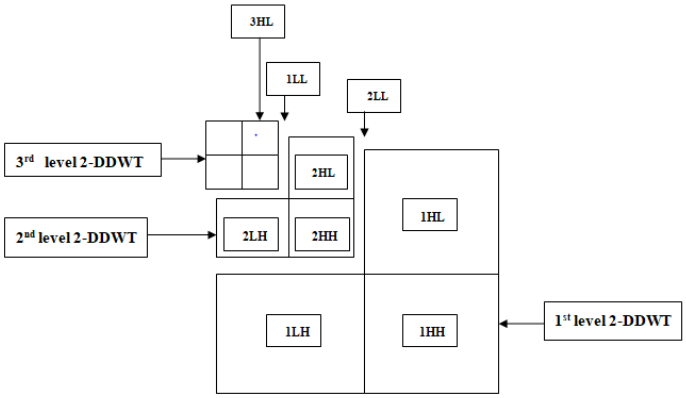


Figure 1: Multi level decomposition framework of DWT [22].

3.2.1 Mathematical Formulation of Haar-DWT

For an input image $P(x, y)$, the Haar wavelet transform [26] decomposes the image into approximation and detail coefficients through low-pass and high-pass filtering followed by down-sampling. The one-dimensional Haar transform for two adjacent pixels $x_1$and x_2 is defined as equation (11) and (12):

$$A = \frac{x_1+x_2}{\sqrt{2}} \quad (11)$$

$$D = \frac{x_1-x_2}{\sqrt{2}} \quad (12)$$

where A represents the low-frequency coefficient and D denotes the detail or high-frequency coefficient.
Applying transformation along picture rows and columns produces shown in equations (13), (14), (15) and (16) into four sub-bands:

$$LL = L_r\big(L_c(P)\big) \quad (13)$$

$$LH = L_r\big(H_c(P)\big) \quad (14)$$

$$HL = H_r\big(L_c(P)\big) \quad (15)$$

$$HH = H_r\big(H_c(P)\big) \quad (16)$$

where L_r and L_c denote low-pass filtering along rows and columns, respectively, and H_r and H_c denote high-pass filtering along rows and columns.
The two-dimensional wavelet decomposition can be represented as shown by equation (17):

$$P(x, y) = LL + LH + HL + HH \quad (17)$$

where the LL sub-band contains the approximation information of the image, while LH, HL, and HH contain vertical, horizontal, and diagonal detail information, respectively.
DWT method in Figure 1 [22] is employed to modify the repeated areas using Haar-DWT, which is the DWT iteration with the fewest steps. A 2-layer Haar DWT comprises of two stages:

Stage 1: Horizontal Operation

a. Divide the image into rows.
b. Apply the Haar transform to each row:
   - Average adjacent pixels (low-pass filter).
   - Calculate the difference between adjacent pixels (high-pass filter).
c. Store the averages on the left and the differences on the right.

Stage 2: Vertical Operation

a. Divide the image into columns.
b. Apply the Haar transform to each column:
- Average adjacent pixels (low-pass filter).
- Calculate the difference between adjacent pixels (high-pass filter).

c. Store the averages at the top and the differences at the bottom.

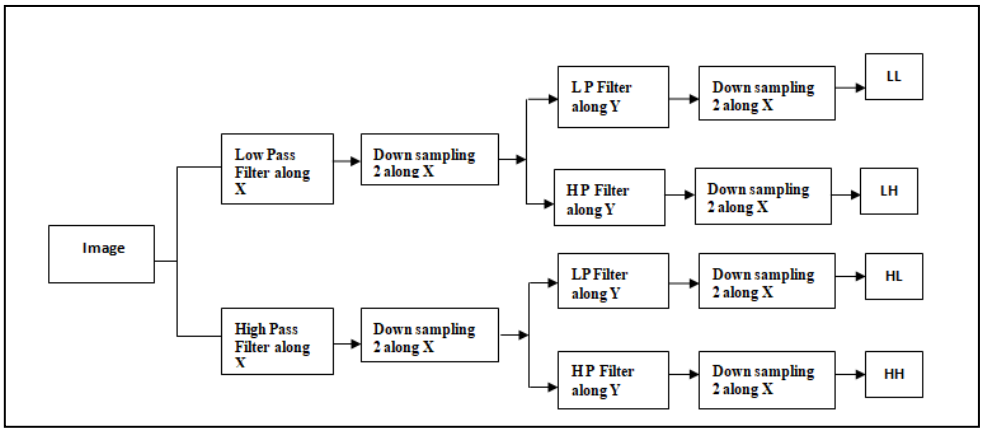


Figure 2: Significant parts (smooth areas) of the spatial domain image are in the DWT's approximation band of low frequency wavelet coefficients.

The Haar-DWT process in Figure 2 yields four sub-bands by following equation (18):

$$W = \begin{bmatrix} LL & HL \\ lH & HH \end{bmatrix} \tag{18}$$

where W represents the transformed wavelet coefficient matrix and:
- LL (Low-Low): Low-frequency component, resembling the original image.
- HL (High-Low): Horizontal high-frequency component.
- LH (Low-High): Vertical high-frequency component.
- HH (High-High): Diagonal high-frequency component.

DWT method, specifically the Haar-DWT, provides an efficient and effective way to decompose images into distinct frequency sub-bands. The 2-layer Haar-DWT process, consisting of horizontal and vertical operations, enables the separation of low-frequency and high-frequency components.

3.3 Discrete Cosine Transform (DCT):

DCT [2-4] algorithm divides an image into segments where the areas of a given image are analyzed according to their importance to the human vision system. This process optimizes image quality. Discrete cosines can then be applied to the image, turning it into another format that allows for the embedding of information within less significant coefficients. This method does not distort the image while encapsulating the data, which is placed into less significant coefficients. While the one-dimensional DCT can be applied to one-dimensional signals such as speech waves, the two-dimensional DCT as per equation (19) is required in analyzing digital images. Where, input image A and output image B, the two-dimensional Discrete Cosine Transform is defined by equation (19) as shown in below expression:

$$B_{pq} = \propto_p \propto_q \sum_{M=0}^{p-1} \sum_{N=0}^{q-1} A_{mn} \quad \cos\frac{\pi(2m+1)p}{2M} \cos\frac{\pi(2n+1)q}{2N} \tag{19}$$

Where:

$$\propto_{\mathrm{p}} = \begin{cases} \dfrac{1}{\sqrt{\mathrm{M}}} & \mathrm{p} = 0 \\ \dfrac{\sqrt{2}}{\sqrt{\mathrm{M}}} & 1 \leq \mathrm{p} \leq \mathrm{M} - 1 \end{cases}$$

and

$$\propto_q = \begin{cases} \frac{1}{\sqrt{N}} & q = 0 \\ \frac{\sqrt{2}}{\sqrt{N}} & 1 \leq q \leq N-1 \end{cases}$$

By combination of hybrid DWT-DCT provides robustness within this framework enhancing the security by reduction of visual distortion. This hybrid method offers a secure and efficient solution for data concealment to increase resistance against steganalysis and data extraction as well as to improve detection resistance boosting data hiding capabilities.

3.3.1 Mid-Frequency Coefficient Selection Theory

The transform coefficients are allotted into low, mid and high frequency resolution after passing through DCT on the approximate sub channel obtained from DWT decomposition. The maximum energy and most visual artifacts are contained in low-frequency coefficients regions, whereas high resolution areas are prone to high sensitivity [27] for compression and attacks. Therefore, the proposed strategy opted only mid-frequency domain of coefficients for secret information embedding which provides an equilibrium among imperceptibility and robustness. The selected coefficient region is defined by equation (20) as the coefficient selection region:

$$\Omega_{MF} = \{C(u,v) \mid T_1 < u + v < T_2\} \quad (20)$$

where (T_1) and (T_2) denotes lower and upper values of threshold which define the embedding areas. The total number of coefficients can be embedded is given by equation (21) as follows:

$$K = \mid \Omega_{MF} \mid \quad (21)$$

where (K) represents available embedding capacity in the transform region. This strategy of coefficients selection decreases visual distortion by aims to maintain resistance against possible common operations related to image processing and statistical steganalysis attacks.

3.3.2 Transform-Domain LSB Embedding Model

After the identification of suitable transform channel coefficients, secret information as picture or text is converted into corresponding binary bit streams before being embedded. Let that payload be denoted as shown in equation (22) Binary payload representation:

$$M = \{m_1, m_2, \ldots, m_L\} \quad (22)$$

where (L) depicted as total number of secret bits. Hence, an adaptive LSB modification [25-26] paradigm is incorporated for concealment of secret payload of information within the DCT coefficients can be expressed as equation (23) follows:

$$c_i' = \begin{cases} c_i, & \text{if } LSB(c_i) = m_i \\ c_i - 1, & \text{if } LSB(c_i) = 1,\ m_i = 0 \\ c_i + 1, & \text{if } LSB(c_i) = 0,\ m_i = 1 \end{cases} \quad (23)$$

where $c_i$and c_i' represents respectively original and modified DCT coefficients. The resulting embedding set of coefficients is represented by equation (24) as follows:

$$C' = \{c_1', c_2', \ldots, c_K'\} \quad (24)$$

This strategy incorporates minimal coefficient alteration [27] and therefore conserves picture visual quality which aims to ensure successfully recovering strategy of the covert information.

3.3.3 Inverse Reconstruction Model

This embedded picture should be reconstructed for obtaining final output stego picture following modification of coefficient. The inverse DCT (IDCT) is applied first for recovery of modified approximation sub channel following equation (25):

$$LL' = IDCT(C') \tag{25}$$

This approximate coefficient band after reconstruction are integrates with remaining wavelet sub bands through Inverse DWT(IDWT) by following equation (26) as given below:

$$R' = IDWT(LL', LH, HL, HH) \tag{26}$$

Thus, at the final stage original red channel is replaced by modified red component which then integrate with unchanged blue and green channel to get the result as shown by equation (27).

$$P_s = \{R', G, B\} \tag{27}$$

3.4 Integration of Blockchain Technology

The specific goals are to achieve high security against various steganalysis attacks as well as enhance embedding capacity while managing security [28]. Ensuring robust data recovery despite transmission errors to develop a blockchain-based metadata storage [29] and authentication system [30].

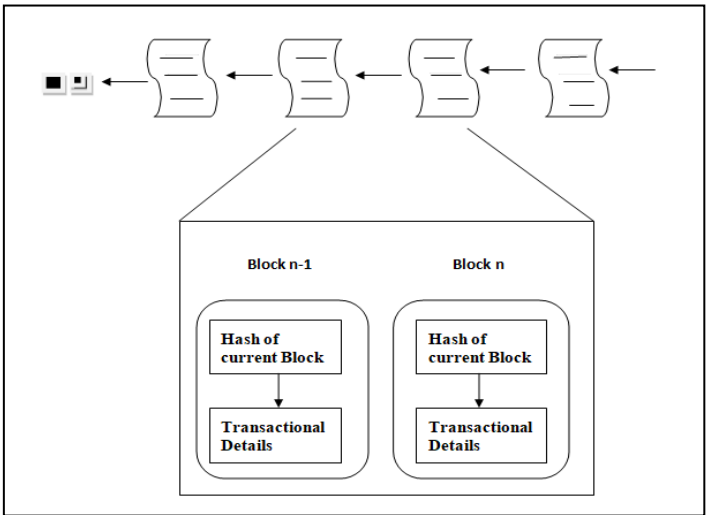


Figure 3: Workflow of Blockchain

The significance of the study contributes to the development of secure as well as efficient image steganography methods, particularly in medical applications. Integration of blockchain [31] in Figure 3, technology ensures the integrity and authenticity of transmitted data, addressing the pressing need for secure medical image transmission. In case of a blockchain, verification [32] of the data is a secure decentralized record.

Cryptography [33] hash function shown in equation (28) for original cover picture and equation (29) for stego picture is used for verification of integrity.

$$H_{image} = SHA256(P) \tag{28}$$

$$H_{stego} = SHA256(S) \quad (29)$$

where P denotes the cover picture, S represents the stego picture, and $SHA256(\cdot)$ is the cryptography hash function [34] produces a 256-bit digest. The generated hashes are stored on the Ethereum blockchain for future verification and tamper detection.

3.4.1 Blockchain Transaction Formation

The metadata transaction submitted to Ethereum is represented as equation (30) [35]:

$$T = \{ID, H_{image}, H_{stego}, TS, P\} \quad (30)$$

Where:

ID= unique picture identifier,
H_{image}= original picture hash,
H_{stego}= stego picture hash,
TS= timestamp,
P= embedding parameters.

The transaction is signed digitally before being proceed for recording on the blockchain. When implementing blockchain [35] with image steganography the data that has been hidden can be verified to confirm that they are real and original. In the blockchain, there is a cryptographic hash function that can be used to check for data manipulation or alteration.

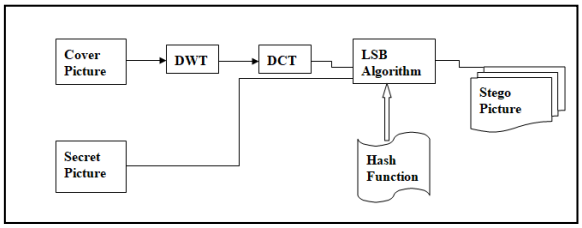


Figure 4: The Embedding Processing using Hash Function

The results of the present study in Figure 4, will contribute to enhancing the approaches to image steganography and avoiding the existing issues to support secure data transmission in various domains.

The complete retrieval and verification sequence is therefore:

Received stego image → DWT–DCT–LSB extraction → recovered secret message → SHA-256 hash computation → blockchain transaction retrieval → stored-hash comparison → blockchain-record validation → authentication decision.

This design ensures that blockchain is used for immutable authentication and integrity verification, while the actual image and secret message remain outside the blockchain.

Information	**Generated from**	**On-chain status**	**Purpose**
I_c	Original cover image	No	Original image remains off-chain
H_c	SHA-256(I_c)	Yes	Reference fingerprint of cover image
I_s	Generated stego image	No	Transmitted/stored outside blockchain
H_s	SHA-256(I_s)	Yes	Integrity verification of stego image
M	Secret message	No	Embedded inside stego image

H_m	SHA-256(M)	Payload-level verification	Detects changes to recovered secret, where enabled
ID	Image/transaction identifier	Yes	Record identification
D	Image dimensions	Yes	Authentication metadata
θ	Embedding parameters	Yes	Verification/audit information
t	Blockchain transaction time	Yes	Timestamp/auditability
Transaction ID	Ethereum transaction	Yes	Immutable retrieval reference

3.4.2 Smart Contract Verification

The authentication of picture is verified by smart contract [36] by comparing stored and calculated hash values using below expression (31):

$$V = \begin{cases} 1, H_{stored} = H_{received} \\ 0, \quad otherwise \end{cases} \quad (31).$$

Where $V = 1$, indicating successful authorization [37] whereas $V = 0$ indicates possible tampering.
This chaining mechanism ensures immutability and prevents unauthorized modification of stored metadata. Integration of blockchain [38] within this framework is for the purpose to ensure indestructible authentication, secure metadata depository, and verifiable data integrity. To implement this realistically, the method execution leverages an independent Ethereum blockchain network. The selection of Ethereum enables customization and its systematic on-chain operations due to mature ecosystem, smart contracts support, and strong development tools such as Remix or IDE etc.

In our implementation:

- For solidity to control metadata depository, validation of image hash, and admittance control policy, smart contracts are involved.
- Web3.py (a Python library) connects the steganographic algorithm with the blockchain backend, allowing real-time interaction with smart contracts. The integrity confidence of the received picture can be expressed as equation (32)

$$A_c = \frac{N_m}{N_t} \quad (32)$$

Where,
N_m= number of matching blockchain records,
N_t= total verification records.
For complete integrity, as shown by expression (33):

$$A_c = 1 \quad (33)$$

Which indicate successful validation of all blockchain.

The IPFS (Interplanetary File System) is optionally used to store encrypted image hashes off-chain, with the IPFS hash anchored to Ethereum for better scalability. Private Ethereum ensures:

- Faster solidarity in comparison with public networks.
- Custom consents for restrained access (unattainable for healthcare and military area) [39].
- Cost-effective operations without reliance on gas tokens.

The smart contract stores:

- A cryptographic hash of the original and stego images (H_{image}, H_{data})[33-34].
- Metadata such as image dimensions, embedding parameters, and timestamp [30, 35].
- Unique identifiers for authentication and auditability [30-31].

3.4.3 Ethereum Block Structure

Each block in Ethereum contains a cryptographic linkage [33] to the previous block by equation (34):

$$B_i = \{Data_i, H(B_{i-1}), Nonce_i\} \quad (34)$$

Where:

- $Data_i$ represents transaction information,
- $H(B_{i-1})$ is the hash of the previous block,
- $Nonce_i$ is the consensus parameter.

The block hash [33] is calculated as expression (35):

$$H(B_i) = SHA256(Data_i \parallel H(B_{i-1}) \parallel Nonce_i) \quad (35)$$

Where concatenation is represented as ||.

3.4.4 Secure Stego-Blockchain Framework

The overall model for secure transmission is represented as equation (36):

$$S_f = \{E(D, C), BC(H, M)\} \quad (36)$$

Where:

- $E(D, C)$ denotes secret embedding information D into cover picture C,
- $BC(H, M)$ represent blockchain storage of hash H and metadata M.

The objective security is expressed as equation (37):

$$max(PSNR + Capacity + A_c) \quad (37)$$

subject to $BER \rightarrow 0$, where $PSNR$ is the Peak Signal-to-Noise Ratio, $Capacity$ represents embed payload, A_c is the authentication confidence, and BER is the Bit Error Rate [40].

3.4.4. End-to-End Architecture of the Proposed Steganography–Blockchain Framework

The proposed framework consists of two data-protection mechanisms, namely, steganography and blockchain. The secret message is embedded into the cover image using the hybrid DWT-DCT-LSB algorithm to obtain the stego image. By the SHA-256 algorithm, the hash values of the original image and stego image are computed. Apart from the hash values of the images, the image ID, size, parameters of embedding process, and the timestamp are registered through an Ethereum smart contract in a cryptographic signature. It is noteworthy that images and the secret message itself are stored outside the blockchain.
On the receiver's side, a similar SHA-256 hash of the retrieved image is computed and compared to the hash stored in the blockchain. In the case of a match, it can be concluded that the image has not been altered; otherwise, unauthorized changes were detected. After the image is validated, the next step is to extract the secret message by applying the same mathematical functions and parameters as in the embedding process. Therefore, the use of steganography and blockchain allows for securing the data-storage process against unauthorized access and malicious tampering. Both methods are used in a sequential manner, and neither of them is considered an integral part of the other. Consequently, the blockchain technology is not thought of as a substitute for the encryption process or steganographic protection.

End-to-end workflow

- Sender: Secret-message embedding
- Cover image + secret message → Red-channel DWT–DCT–LSB embedding → Stego image
- Blockchain registration
- Compute SHA-256 hashes → Register image hashes and metadata through Ethereum smart contract
- Transmission and retrieval
- Transmit stego image and provide transaction reference to the receiver
- Receiver: Integrity verification
- Recompute stego-image hash → Retrieve registered hash → Compare hashes
- Hash match
- Proceed to message extraction

- Hash mismatch
- Flag possible image modification

3.4.5 Payload Capacity Model

Measurement of the embedding capability in bits per pixels (bpp) of proposed strategy shown in equation (38) as follows:

$$C_{bpp} = \frac{L}{M \times N} \tag{38}$$

where (L) denotes the payload to be embedded and $M \times N$ represents the picture resolution. The calculation of payload percentage is formulated by equation (39) as follows (Payload percentage):

$$P_c = \frac{L}{M \times N} \times 100 \tag{39}$$

The objective of evaluation of this proposed paradigm enables by formulations which quantify the trade-off among embedding capacity and visual quality.

The proposed method meets the current requirement of safe data transfer in applications where information security is of utmost importance. Blockchain ensures [41] the validity of the data being transmitted and hybrid transformation ensures its security from third-party intrusion. The first goal is to propose a secure and high-performance red image steganography method to create a new steganography algorithm based on the combined approach to encode the secret data in the red colour images [8]. Integrating blockchain technology [42] utilizes to store metadata and authenticate the embedded data, providing tamper-proof and decentralized security which assesses performance of proposed method in terms of:

- Security: Resistance to steganalysis and attacks.
- Capacity: Embedding capacity and data hiding efficiency.
- Robustness: Data recovery integrity during transmission and processing.

3.5 CNN Based Steganalysis

Evaluation for security of the proposed red-channel DWT–DCT–LSB strategy against modern learning-based detectors, a Convolutional Neural Network (CNN) based steganalysis detector was incorporated as an external adversarial classifier [43]. This is employed not for component of the embedding or extraction process; rather, it is used entirely for assessing whether hidden data can be detected from the generated stego images [44]. The process is depicted through following equations. Let us consider the cover picture is denoted by P_c and stego picture as P_s

The CNN detector is represented by parameter expression (40) below:

$$f_\theta(I) \tag{40}$$

Where:

$$f_\theta(I) = \begin{cases} 0, & \text{Cover} \\ 1, & \text{Stego} \end{cases}$$

The classification accuracy is measured by equation (41) as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{41}$$

and the AUC score is measured by equation (42):

$$AUC = \int_0^1 TPR(FPR)\, d(FPR) \tag{42}$$

For a highly secure steganographic system, $Accuracy \approx 50\%$ and $AUC \approx 0.5$ which indicates random guessing behavior.